

OSTROV PREOBRAZENIJ, Russian Federation

WMO: 215040

Lat: 74.667N

Lon: 112.933E

Elev: 57

StdP: 100.64

Time Zone: 9.00 (E09)

Period: 82-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.2	-39.2	-43.0	0.1	-41.0	-41.0	0.1	-39.0	16.9	-24.2	13.8	-26.1	3.6	180	1.215

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	12.1	9.8	9.3	7.6	7.1	5.9	10.0	12.0	7.6	9.1	6.0	7.1	5.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.0	11.2	6.5	6.0	8.3	5.0	5.4	6.6	29.7	12.0	24.3	9.3	20.9	7.1	20.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.3	13.3	11.9	DB	-42.2	18.4	2.2	3.5	-43.8	20.9	-45.1	23.0	-46.4	24.9	-48.0	27.5	(4)
			WB	-42.3	14.5	2.2	3.1	-43.8	16.7	-45.1	18.5	-46.3	20.2	-47.9	22.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	-14.2	-30.1	-28.8	-26.7	-20.2	-9.6	-0.6	3.4	2.8	-0.9	-11.3	-23.1	-26.1	
	DBStd	13.36	5.91	5.66	5.62	5.48	3.94	3.57	2.96	3.14	3.15	5.92	4.56	5.76	
	HDD10.0	8837	1242	1088	1138	906	608	319	208	225	328	662	994	1118	
	HDD18.3	11873	1500	1321	1397	1156	866	568	464	482	578	920	1244	1377	
	CDD10.0	5	0	0	0	0	0	1	3	2	0	0	0	0	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	5.8	4.9	5.3	5.5	6.4	6.2	6.1	5.9	6.3	6.1	6.4	5.5	5.3	
Precipitation	PrecAvg	281	17	12	15	16	24	21	30	40	36	33	22	17	
	PrecMax	395	54	28	26	41	47	43	64	80	86	58	44	39	
	PrecMin	163	3	1	2	0	4	5	10	16	10	11	8	7	
	PrecStd	51	11	6	6	9	9	9	13	18	19	10	10	7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.7	-10.7	-8.1	-4.9	0.1	12.5	20.0	14.8	7.7	0.1	-9.4	-8.5	
		MCWB	-13.1	-11.1	-9.1	-5.7	-0.6	9.3	16.1	12.0	6.6	-0.1	-9.9	-8.7	
	2%	DB	-15.8	-15.9	-12.9	-8.1	-1.1	8.3	13.0	11.7	5.6	-1.2	-12.0	-13.7	
		MCWB	-16.2	-16.2	-13.2	-8.6	-1.6	6.5	10.5	9.7	5.1	-1.7	-12.4	-14.0	
	5%	DB	-18.5	-18.3	-16.0	-10.4	-2.5	4.9	9.8	9.0	4.1	-2.2	-14.5	-16.1	
		MCWB	-18.7	-18.6	-16.3	-10.9	-3.1	3.6	8.1	7.6	3.7	-2.7	-14.9	-16.4	
	10%	DB	-21.1	-20.7	-18.9	-12.7	-3.9	3.1	7.2	7.1	2.9	-3.2	-16.7	-18.0	
		MCWB	-21.3	-20.9	-19.1	-13.2	-4.5	2.1	5.8	6.0	2.5	-3.6	-17.0	-18.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-13.1	-11.0	-8.8	-5.5	-0.4	9.1	15.7	12.2	6.5	0.0	-9.8	-8.8
			MCDB	-12.7	-10.6	-8.2	-4.7	-0.1	11.7	19.3	14.1	7.1	0.1	-9.5	-8.1
2%		WB	-16.2	-16.3	-13.4	-8.5	-1.6	6.3	10.6	10.0	5.2	-1.5	-12.3	-13.9	
		MCDB	-15.9	-16.1	-13.0	-8.1	-1.2	8.1	12.6	11.7	5.6	-1.3	-12.1	-13.7	
5%		WB	-18.7	-18.7	-16.4	-10.7	-3.0	3.6	8.0	7.7	3.8	-2.6	-14.9	-16.4	
		MCDB	-18.5	-18.4	-16.1	-10.3	-2.5	4.6	9.8	8.9	4.1	-2.2	-14.6	-16.2	
10%		WB	-21.3	-21.0	-19.2	-13.1	-4.4	2.2	6.0	6.2	2.6	-3.7	-17.0	-18.3	
		MCDB	-21.1	-20.8	-18.9	-12.7	-3.8	2.9	7.2	7.1	2.9	-3.3	-16.7	-18.0	
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.5	6.0	5.8	4.1	3.4	4.6	3.5	2.6	4.0	4.4	5.0	
		MCDBR	8.3	7.9	9.3	8.2	4.7	7.2	10.1	7.5	3.7	4.2	6.2	6.4	
		MCWBR	8.2	7.9	9.2	8.1	4.4	5.4	7.8	5.7	3.4	4.4	6.2	6.2	
	5% WB	MCDBR	8.3	8.0	9.1	8.3	4.5	6.8	9.6	7.4	3.7	4.3	6.2	6.3	
		MCWBR	8.1	7.9	9.0	8.1	4.3	5.3	7.8	5.9	3.5	4.5	6.1	6.2	
Clear-Sky Solar Irradiance	taub	N/A	0.172	0.267	0.300	0.324	0.326	0.361	0.349	0.287	N/A	N/A	N/A		
	taud	N/A	1.873	2.006	2.054	2.078	2.262	2.321	2.390	2.407	N/A	N/A	N/A		
	Ebn at Noon	0	289	618	772	814	837	783	734	651	N/A	0	0		
	Edh at Noon	0	19	69	103	119	104	95	77	49	N/A	0	0		
All-Sky Solar Radiation	RadAvg	0.00	0.10	1.37	3.79	5.64	6.16	4.95	2.79	1.15	0.23	0.00	0.00		
	RadStd	0.00	0.01	0.10	0.22	0.29	0.51	0.47	0.33	0.11	0.03	0.00	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page